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Physics HL Internal Assessment

Magnetic Braking through Eddy Currents

What is the relationship between the horizontal separation distance (m) between a stationary aluminium sheet and a N35 neodymium magnet and the decay constant (s^{-1}) of a swinging pendulum?

Candidate code: krp114

Word Count: 2813

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1. Introduction

When an object wants to decelerate, a braking force must be applied, generally done through frictional force. However, this generates unwanted thermal energy which wears the braking system overtime by putting unnecessary stress. To avoid this, magnetic braking can be utilised where braking is done through electromagnetic induction. In oscillatory systems, damping occurs due to loss of mechanical energy. If a magnetic system undergoes harmonic motion, non-ferromagnetic materials found near the arrangement would cause generation of an opposing magnetic field which damps the system. This experiment aims to investigate the relationship between distance between the pendulum and aluminium and the decay constant of the system.

2. Research Question

What is the relationship between the horizontal separation distance (m) between a stationary aluminium sheet and a N35 neodymium magnet and the decay constant (s^{-1}) of a swinging pendulum?

3. Background Information

3.1. Simple harmonic motion

Simple Harmonic Motion (SHM)[1] is a concept used in classical mechanics that describe the periodic motion of an object about an equilibrium position. It is initially induced by a disturbance in the system however is sustained through the relationship between is displacement from the mean position and its acceleration vector.

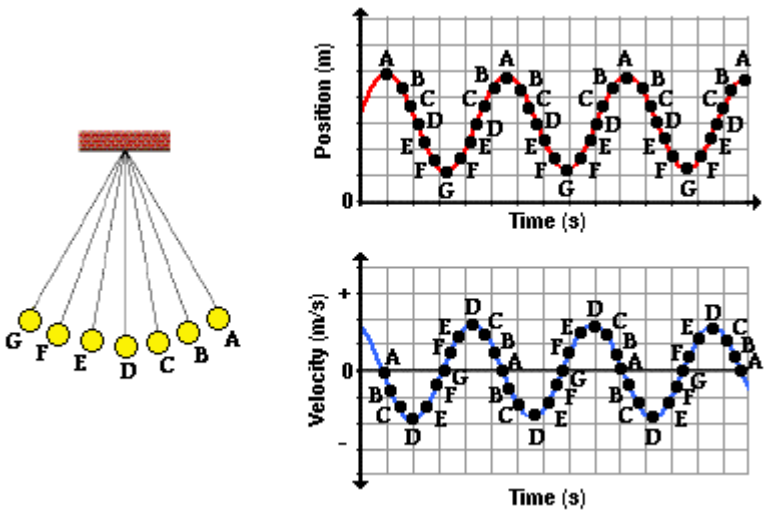


Figure 1 - Harmonic motion and its motion graphs

[1] "Physics Tutorial: Pendulum Motion." *The Physics Classroom*, www.physicsclassroom.com/class/waves/lesson-0/pendulum-motion. Accessed 15 Jan. 2026.

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This leads to the fundamental equation, defining any oscillatory motion –

$$a = -\omega^2 x \quad (1)$$

Where:

a = acceleration of object (ms^{-2})

x = displacement of object from its mean position (m)

ω = angular frequency of object (rad s^{-1})

However, empirically energy isn't conserved in the system and is lost in the form of air resistance and friction.

3.2. Electromagnetic Induction

Electromagnetic induction is a natural phenomenon where varying magnetic flux linkage induces a current in any conducting material. This relation is governed by the Faraday's law as given through the equation –

$$\varepsilon = -\frac{d\varphi}{dt} \quad (2)$$

Where φ is the magnetic flux, the number of intersecting field lines and $\frac{d\varphi}{dt}$ shows the rate of change of magnetic flux linkage. Magnetic flux can also be writing as –

$$\varphi = BA\cos\theta \quad (3)$$

$$\varphi \propto B$$

Lenz law states that eddy currents - the induced currents, produce an opposing magnetic field, which opposes the direction of motion, in this case being the motion of the oscillating magnet.

[2] Ling, Samuel J., et al. "13.1 Faraday's Law - University Physics Volume 2." *OpenStax*, openstax.org/books/university-physics-volume-2/pages/13-1-faradays-law. Accessed 19 Mar. 2026.

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With greater number of fields line intersecting over time, a greater electromotive force will be generated. The generated voltage is dictated by the local movement of the charge relative to the magnetic field penetrating the conductor.

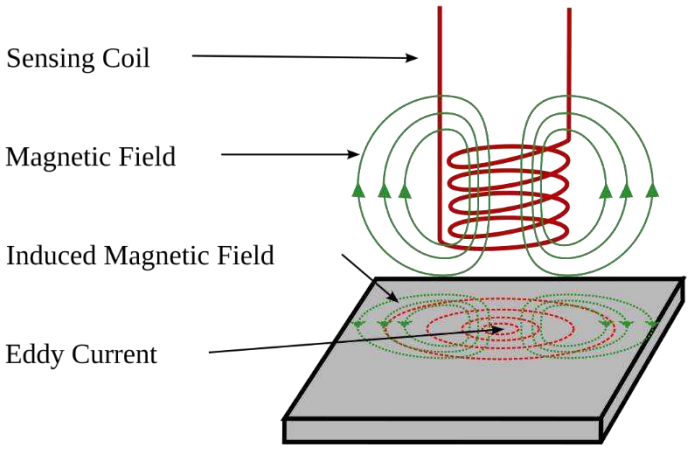


Figure 2 - Generation of eddy currents

3.3. Eddy current damping

As the magnet oscillates, eddy currents generate in the aluminium sheets. If the distance between the pendulum and sheet changed, the magnetic field strength (B) varies sharpy since–

$$B \propto \frac{1}{d^3}$$ [3]

The electromotive force can be converted to –

$$\varepsilon = -\frac{d\varphi}{dx}v$$

Through conservation of energy, loss of mechanical energy equates to gain of electrical energy.

$$bv^2 = \frac{\left(\frac{d\varphi}{dx}v\right)^2}{R}$$

b represents the damping coefficient; the two different forms of power are equated to relate the damping coefficient and the distance.

$$b = \frac{1}{R}\left(\frac{d\varphi}{dx}\right)^2 \quad (4)$$

$$\because \frac{d\varphi}{dx} \propto \frac{1}{d^3}$$

[3] Ling, Samuel J., et al. "12.4 Magnetic Field of a Current Loop - University Physics Volume 2." *OpenStax*, openstax.org/books/university-physics-volume-2/pages/12-4-magnetic-field-of-a-current-loop. Accessed 19 Mar. 2026.

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$$\boldsymbol{b \propto \frac{1}{d^6}}\ ^{4} \qquad (5)$$

In damped harmonic oscillators, the decay in amplitude it modelled through –

$$A = A_0 e^{-\lambda t}$$

Where λ is the decay constant, the rate at which amplitude decays. λ can be related to the damping coefficient as such-

$$\lambda = \frac{b}{2m_{magnet}} \qquad (6)$$

Equation (5) and (6) can be related to form the relationship

$$\boldsymbol{\lambda \propto \frac{1}{d^6}} \qquad (7)$$

This illustrates the fact that increasing distance will lead to decreasing decay constant.

4. Hypothesis –

As the distance (d) between the pendulum and aluminium plates increase, decay constant of the pendulum decreases significantly. This is because distance and magnetic flux linkage are related through an inverse power relationship. This leads to weaker magnetic flux linkage, and subsequently induced eddy currents, which reduces damping effects.

5. Variables

- **Independent variable** – distance between the magnet and aluminium plate (mm)

The distance will be varied over 10 intervals – 50, 60, 70, 80, 90, 100, 110, 120, 130, 140 mm. A vernier calliper will be used to change accurately measure the distances. Thus, this will be changed to observe its effect on oscillations half-life.

- **Dependent Variable** – the decay constant of the pendulum system (s^{-1})

This will be calculated through the time taken for the oscillations to decay to half its initial amplitude and relating through the equation –

[4] Wiederick, H. D., et al. "Magnetic Braking: Simple Theory and Experiment." *American Journal of Physics*, vol. 55, no. 6, 1987, pp. 500–503.

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$$\gamma = \frac{\ln(2)}{T_{\frac{1}{2}}} \quad (8)$$

The time taken will be measured through a video analysis software known as Tracker (±0.02s). This will ensure precise measurement of the oscillations. To ensure reliability, the time period will be averaged over 10 trials.

- **Control Variable**

I. Initial angle of release (θ) –

Angle of release dictates the initial amplitude of the oscillation. If not controlled, the half-life will be measured variably, reducing accuracy of the results. A fixed protractor should be used along with a starting mark, where all experimentation should begin.

II. Mass of magnet (kg) –

The damping coefficient is proportional to the mass of the magnet. Varying masses will therefore lead to differing decay constants. The same experimental setup of magnet and string should be used. Verify the weight of the magnet before and after experimentation to certify constant mass.

III. Thickness of aluminium plate (m) –

The magnitude of eddy currents generated will increase since magnetic flux linkage increases. This will increase the damping effects. The same aluminium plate will be used throughout all trials.

IV. Temperature of aluminium plate (^{0}C) –

Eddy currents raise the temperature of plate over time. At increased temperature, the aluminium plate will worsen as a conductor, increasing its resistance. This reduces the damping effect over the course of each trial. Between each trial, allow the plate to cool down for 30 seconds. Ensure the ambient room temperature remains constant.

6. Apparatus used

Instruments & uncertainties	Additional Information
Measurement tape (±0.0005 m)	-

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Neodymium disk magnet	The magnet used is N35 graded, with strength approximately 1.19 T. The disk is 30mm in diameter and 3mm thick.
Lightweight string	-
Aluminium sheet – 2mm	The sheet must be large enough to cover the entire range of motion of the pendulum, to ensure constant damping throughout.
Clamp	-
Protractor ($\pm 0.5^0$)	-
Smartphone camera – 120 fps (±0.00208 s)	-
Additional masses (±0.001 kg)	The neodymium magnet may not be able to perform SHM. Adding additional mass will increase inertia and allows oscillation. However, the mass must not be metallic, as that might affect interactions between magnet and plate.

Table 1 - Apparatus required

7. Procedure

7.1. Experimental setup

a) To an inextensible, lightweight string, attach the N35 disk magnet using cello tape. Attach the string to a retort stand. Ensure that the stand is vertically upright, with sufficient space for oscillations.
b) Use a vernier calliper to measure the exact intervals for distance and place the aluminium plate using non-magnetic blocks as support. Make sure that the plate is vertically upright.
c) Place a 30cm ruler vertically. Since it is used for calibration of the tracker software, the ruler must be kept upright with clear markings. Place a highspeed camera 0.5 meters away from the experimental setup, while making sure that the pendulum setup is centred in its frame. Recording in high resolution (1080p – 4K) allows markings and displacements to be accurately computed by software.
d) Using a protractor to measure $\theta = 20^{\circ}$. Displace the pendulum by θ and mark the point (A_0) right below the magnet. This serves as the baseline initial amplitudes for all

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subsequent trials. Further, measure the distance between the equilibrium and A_0. Mark the midpoint to identify when amplitude decays to half its initial value.

7.2. Data collection

e) Without placing the plate, complete a controlled trial. Record and measure time taken for $T_{1/2}$. This allows calculation of decay through air resistance.
f) For the independent variable, set the aluminium plate accordingly. Measurements must stem from the face of the magnet to the plate.
g) Begin the recording and displace the magnet to the marking. Leave the magnet, allowing it to oscillate without providing any additional force.
h) During the trial, make sure that the oscillations remain parallel to the plate. If the oscillations are skewed, the distance will be inconsistent between each amplitude.
i) Continue the recording until the amplitudes decay to $A_{1/2}$.
j) Wait 30 seconds and repeat the trials 10 times for the same interval.
k) Change the independent variable, interval of distance and repeat steps (e) to (h)

- **Risk Assessment**

Type of hazard	Description of hazards	Prevention
Experimental hazards	Neodymium magnets are extremely strong. If handled carelessly, may cause the magnet to snap onto skin and injure it or may be fragmented upon impact.	Ensure that the magnets are handled with gloves and with a firm grip. Keep all ferrous objects at least 0.5 meters away from magnet.
	Aluminium maybe but extremely sharp. This may cause injury to skin.	Cover sharp edges with any soft material such as foam.
Ethical consideration	Neodymium magnet can disrupt magnetic technologies such as bank cards or laptops.	Keep all electronic devices away from the setup to avoid issues.

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Environmental hazards	Neodymium is a rare earth metal that when mined, causes production of mine tailing.[5]	The magnet will be used from previous experimentations and will be passed onto the laboratory, ensuring that the magnet remains magnetisation.
	Aluminium is a resource that should not be thrown after use.	Aluminium may be passed along for possible future usage.

Table 2 - Experimental, Ethical and environmental considerations

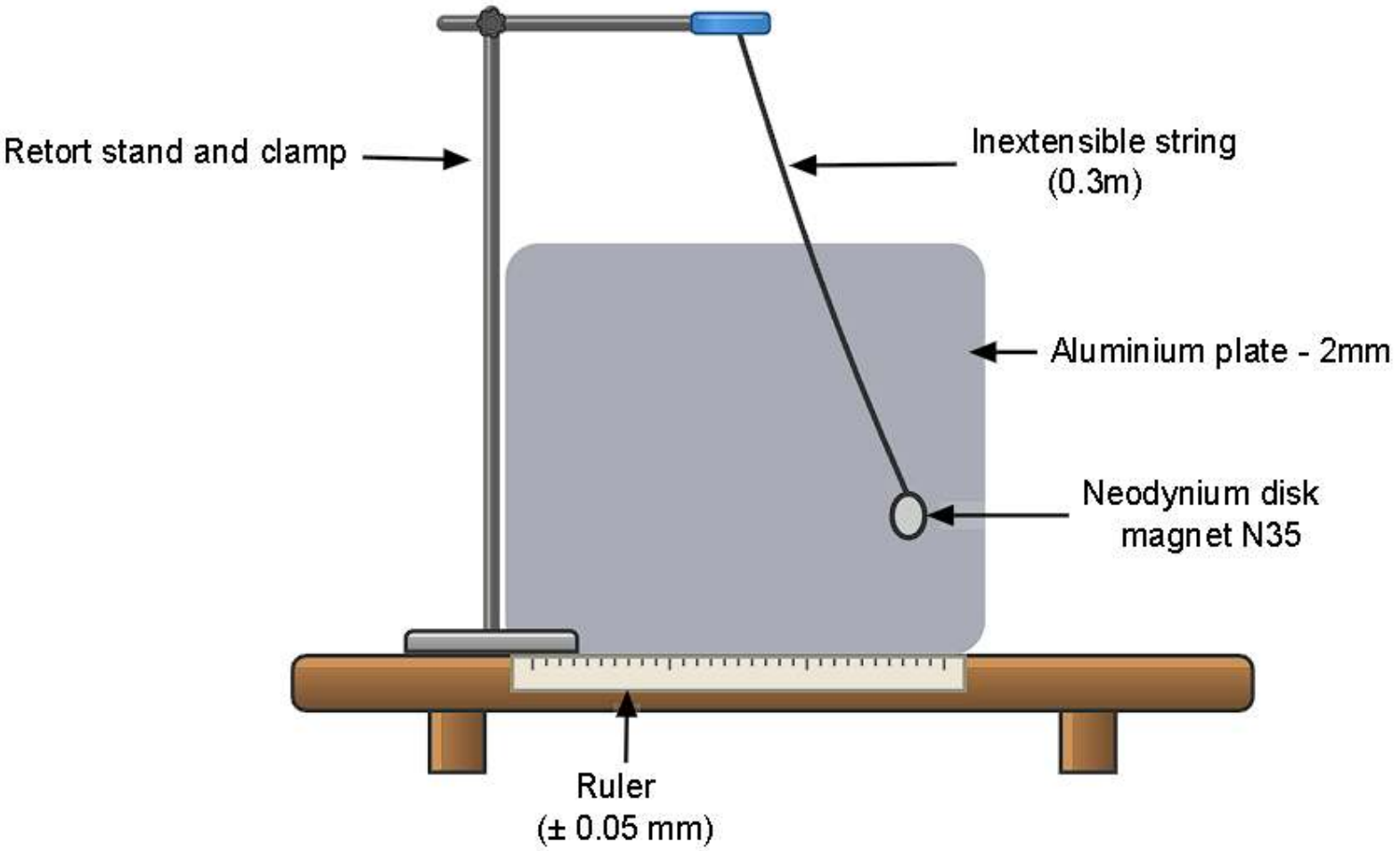


Figure 3 - Frontal view of experimental setup

[5] Abigail Martin, Alastair Iles. "The Ethics of Rare Earth Elements over Time and Space." *HYLE 26-1 (2020): The Ethics of Rare Earth Elements Over Time and Space*, hyle.org/journal/issues/26-1/martin.htm. Accessed 16 Jan. 2026.

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8. Data Analysis

8.1. Constants

Constant	Measured value
Mass of the magnet (m)	0.05 kgs
Length of pendulum (L)	0.3 m
Angle of release (θ)	20°
Aluminium Thickness (δ)	2 mm
Half amplitude ($A_{1/2}$)	0.05m

Table 3 - Controlled variable's values

8.2. Raw data collection

Trial	Half-life of the pendulum (s) (±0.02 s)
1	293.10
2	289.70
3	286.00
4	312.40
5	292.70
6	294.20
7	295.20
8	305.40
9	314.30
10	304.70

Table 4 - Controlled Trial / No plate

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Distance between magnet and plate (d)(m) (±0.5mm)	Half-life of the pendulum as measured by "Tracker" ($T_{1/2}$) (s) (±0.02 s)									
	Trial 1	**Trial 2**	**Trial 3**	**Trial 4**	**Trial 5**	**Trial 6**	**Trial 7**	**Trial 8**	**Trial 9**	**Trial 10**
50.0	155.00	148.90	159.70	157.60	166.80	159.60	152.50	147.60	161.60	155.70
60.0	217.40	212.70	200.60	225.90	222.20	219.90	223.80	207.10	235.20	215.20
70.0	258.00	261.90	226.60	271.40	254.90	237.10	250.70	282.10	259.10	239.90
80.0	272.30	278.30	267.40	262.60	287.00	283.20	274.00	273.30	275.70	269.30
90.0	284.40	286.30	301.50	271.40	295.40	303.80	274.90	286.80	287.60	282.90
100.0	285.00	279.00	264.90	297.30	252.50	296.50	301.20	278.60	297.10	298.90
110.0	302.40	288.70	298.10	296.40	284.50	287.80	301.60	287.60	299.30	295.10
120.0	293.20	297.40	300.00	296.90	298.20	304.20	294.10	298.60	293.80	300.90
130.0	295.90	305.90	302.60	280.80	301.00	304.20	295.40	288.10	300.10	303.60
140.0	290.70	306.50	318.40	288.90	294.20	318.50	297.10	295.40	303.40	310.70

Table 5 - Raw data of half-lives against distance

8.3. Data processing

8.3.1 Controlled Trial

Initially, the controlled data set will be analysed, which will form the basis for damping caused by air resistance.

The mean of the controlled trial is as follows -

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$$T_{\frac{1}{2}} = \frac{293.1 + 289.7 + 286.0 + 312.4 + 292.7 + 294.2 + 295.2 + 305.4 + 314.3 + 304.7}{10}$$

$$T_{\frac{1}{2}} = 300.8\ s$$

To calculate standard deviation, the following formula can be utilised-

$$\sigma = \sqrt{\frac{\sum(t_i - t_\mu)^2}{n-1}}$$

$$\sqrt{\frac{(293.1 - 300.8)^2 + (289.7 - 300.8)^2 + (286.0 - 300.8)^2 \ldots + (304.7 - 300.8)^2}{9}}$$

$$\sigma = 12.7\ s$$

Standard error can be calculated using the formula –

$$\sigma = \frac{\sigma}{\sqrt{N}}$$

$$\sigma = \frac{12.7}{\sqrt{10}} = 4.0\ \boldsymbol{s}$$

$$\therefore \boldsymbol{T_{control} = 300.8 \pm 4.0\ seconds}$$

The decay constant is given through –

$$\lambda_{air\ resistance} = \frac{\ln(2)}{T_{\frac{1}{2}}}$$

$$\therefore \lambda_{air\ reistance} = \frac{\ln(2)}{300.8} = 0.002304\ s^{-1}$$

Since, the uncertainty in time is equal to uncertainty in decay constant –

$$\frac{\Delta\lambda}{\lambda} = \frac{\Delta T}{T}$$

$$\Delta\lambda = 0.000031\ s^{-1}$$

$$\therefore \boldsymbol{\lambda_{air\ resistance} = (2.304 \pm 0.031) \times 10^{-3} s^{-1}}$$

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To calculate the margin of error, or the standard error in this experiment, the following formula can be utilised –

$$\sigma = \frac{\sigma}{\sqrt{N}}$$

$$\sigma = \frac{5.86}{\sqrt{10}} = \mathbf{1.853\ s}$$

8.3.2 Sample Calculation

Mean time taken for oscillation to reach 50% its initial value –

$$T_{\mu} = \frac{155.0 + 148.9 + 159.7 + 157.6 + 166.8 + 159.6 + 152.5 + 147.6 + 161.6 + 155.7}{10}$$

$$T_{\mu} = 156.5$$

$$\therefore \boldsymbol{T_{50mm} = 156.5}$$

Uncertainty in half life is dependent on two factors, uncertainty in high-speed camera's frame rate and standard error of the sample. A root sum square approach can be used, to quantify combined uncertainty.

$$\Delta T_{tracker} = \pm 0.00208\ s$$

Standard deviation –

$$\therefore \sigma = \sqrt{\frac{(155.0 - 156.5)^2 + (148.9 - 156.5)^2 + (159.7 - 156.5)^2 \ldots + (155.7 - 156.5)^2}{9}}$$

$$\sigma = 5.859654\ \text{s}$$

$$\boldsymbol{\sigma \approx 5.86\ s}$$

Thus, the uncertainty in half life is –

$$\Delta T = \sqrt{(0.00208)^2 + (1.853)^2}$$

$$\Delta T = 1.853$$

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$$\boldsymbol{T_{50mm} = 156.5 \pm 1.9\ s}$$

This proves that uncertainty stems from physical variance rather than experimental errors.

To calculate decay due to eddy current braking, the sum decay constant's difference with decay constant due to air, will yield the value.

$$\lambda_{total} = \frac{\ln(2)}{156.5} \approx 0.00443\ s^{-1}$$

$$\frac{\Delta\lambda_{total}}{\lambda_{total}} = \frac{\Delta T}{T}$$

$$\therefore \Delta\lambda_{total} = 0.00443 \times \frac{1.9}{156.5}$$

$$\Delta\lambda_{total} = 5.4 \times 10^{-5}\ s^{-1}$$

$$\boldsymbol{\lambda_{total} = (4.428 \pm 0.054) \times 10^{-3}\ s^{-1}}$$

Since $\lambda_{total} = \lambda_{magnetic} + \lambda_{air}$

$$\lambda_{magnetic} = 0.004428 - 0.002304 = \boldsymbol{0.002124\ s^{-1}}$$

$$\frac{\Delta\lambda_{magnetic}}{\lambda_{magnetic}} = \frac{\Delta\lambda_{\text{total}}}{\lambda_{total}} + \frac{\Delta\lambda_{\text{air}}}{\lambda_{air}}$$

$$\Delta\lambda_{magnetic} = \left(\frac{0.054}{4.428} + \frac{0.031}{2.304}\right) \times 2.124 \times 10^{-3}$$

$$\boldsymbol{\Delta\lambda_{magnetic} = 5.448 \times 10^{-5} s^{-1}}$$

Distance (mm) (±0.5 mm)	**Mean $T_{1/2}$ (s)**	**Uncertainty in $T_{1/2}$ (s) ($\Delta T_{1/2}$)**	**Magnetic braking decay constant $\lambda_{magnetic}$ (10^{-3} s^{-1})**	**Uncertainty in $\lambda_{magnetic}$ ($\Delta\lambda_{magnetic}$) ($\pm 10^{-3}$ s^{-1})**
50.00	156.5	±1.85	2.124	0.061

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60.00	218.0	±3.11	0.875	0.055
70.00	254.17	±5.21	0.422	0.064
80.00	274.31	±2.29	0.222	0.037
90.00	287.50	±3.29	0.106	0.041
100.00	285.50	±5.19	0.127	0.054
110.00	294.15	±2.05	0.052	0.035
120.00	297.73	±1.10	0.024	0.032
130.00	297.76	±2.52	0.023	0.037
140.00	302.38	±3.44	-	-

Table 6 - Processed Data Table

$\log_{10}(d/mm)$	$\log_{10}(\lambda/10^{-3}\ s^{-1})$	$\delta \log_{10} \lambda$
1.699	0.327	0.011
1.778	-0.058	0.027
1.845	-0.375	0.066
1.903	-0.654	0.072
1.954	-0.975	0.168
2.000	-0.896	0.185
2.041	-1.284	0.292
2.079	-1.620	0.578
2.114	-1.638	0.699

Table 7 - Logarithmic relation

9. Discussion of results

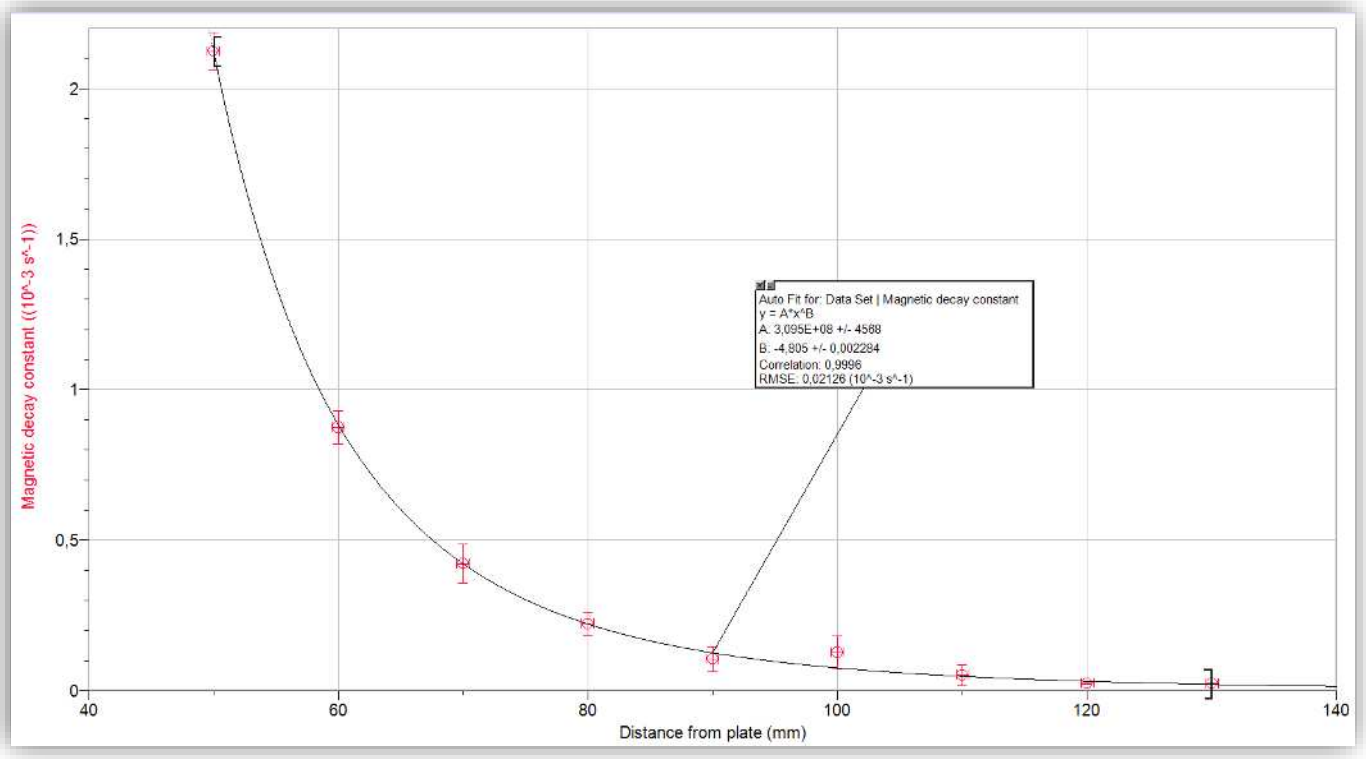


Figure 4 - Plot between distance and magnetic decay constant

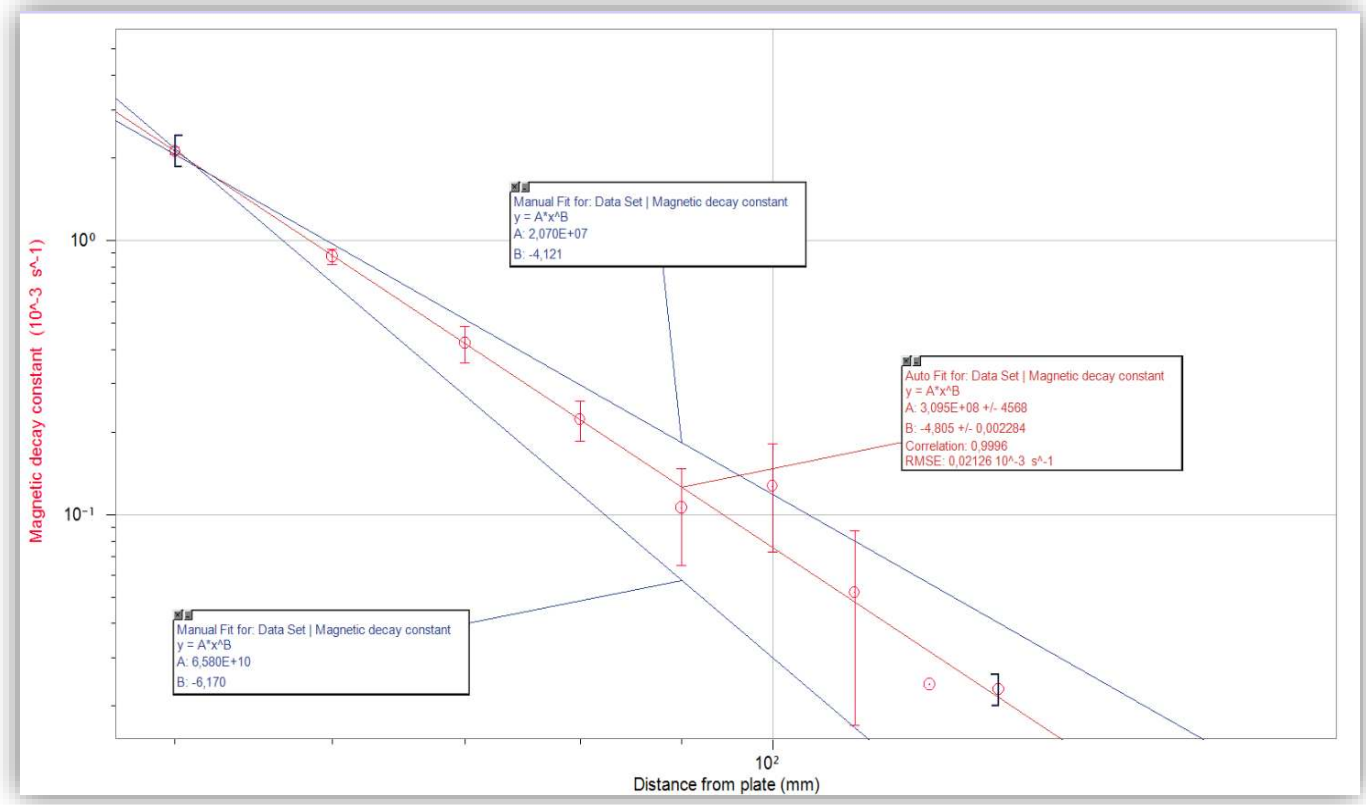


Figure 5 - Log plot between distance and decay constant

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Figure 4 highlights the relationship between distance and decay constant. The data can be modelled through the equation –

$$y = (3.1 \times 10^8)\, x^{-4.805}$$

The correlation of the model is 0.9996, which implies that change in distance accounted for 99.96% of the variation in magnetic decay constant. The initial control trial enabled calculation of decay constant only due to eddy currents which lead to a much more accurate model. The model doesn't follow a linear relation, but a power law. Between 50 mm < d < 80mm, the damping dependence on field is the greatest, seeing a drop of 90%. As distance furthers, the decay constant approaches 0, where at d =140 mm, the total decay is the same as the decay due to air resistance. This occurs since magnetic flux decreases exponentially, as distance increases, leading to generation of weaker eddy currents.

The distances can be classified into 3 main regimes, through its relation to the magnet's radius. Between 50 mm < d <70 mm, the distances are similar to the magnet's radius. The magnetic field exhibits complex behaviour with a strong eddy current, leading to high damping. Between 80 mm < d < 110 mm, the distances are approximately 3× the radius. At such distances, the transition between near sided and far sided magnetic effects take place[6]. This is exemplified by the outlier at d = 100mm, despite overall low variance figures.

Figure 5 linearises the relationship through the use of logarithmic axes. It's important to note, uncertainties for d > 120mm are not included. This is because the uncertainties yielded negative values for the decay constant, which cannot be displayed on the logarithmic axes. Additionally, uncertainties at higher distances distort and appear larger due to the logarithmic axis.

Proportionality constant

The proportionality constant **of 3.1×10^8** signifies the strength of the magnetic damping effects. This is specific to setup, and cannot be predicted through theory. This constant varies drastically with changes in factors such as magnetic field strength, plate thickness, or length of pendulum.

Exponential term

The exponential term is **-4.805**. When compared to theoretical assumption, where $n \approx 6$, the variation is about 20%. This indicates that empirically, damping drops off slower than expected

[6] Levin, Yan, et al. "Magnetic Braking Revisited." *American Journal of Physics*, vol. 74, no. 9, 2006, pp. 815–817.

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from theory. This arises from factors such as edge effects, where the magnetic fields extend beyond the plates, or a deviated pendulum path, leading damping to be stronger at some amplitudes.[7] At small distances, the dipole approximation deviates significantly, so the near field effects reduce the exponent from 6 to 4.8.

Uncertainty Analysis

The uncertainty in distance remains constant throughout at ±0.05mm, as recorded by a measuring ruler. The uncertainty in decay constant remains concentrated towards the mean and ranges between ±0.061 s^{-1} and ±0.032 s^{-1}. This variance in uncertainty stems from experimental error which leads to varying standard error. The uncertainty is greatest when distance is smaller. This is caused by the smaller measurement window since time period shortens. Furthermore, if the path of the pendulum deviates, the uneven swing affects generation of eddy current in non-ideal forms. As distance increases, the damping effects decrease. This allows for a longer measurement period, where the physical environments variation has lesser of an effect. Relative uncertainty however increases with distance, since it's harder to discern between decay from air resistance and magnetic braking.

The furthest distance interval (d = 140mm) happened to display a negative decay constant. This is inherently incorrect as a negative constant implies that energy is being added to the system via eddy currents. However, being within uncertainty error, the decay constant due to magnetic braking is about 0. Thus, at further distances, the eddy currents cannot generate a sufficient enough magnetic field, to cause damping.

10. Conclusion

The aim of this investigation was to examine the relationship between distance between a neodymium magnet and the decay constant of the swinging magnet. The hypothesis was tested to be true as the experiment yielded a negative correlation where the decay constant decreased from 2.124×10^{-3} s^{-1} to 0.023 s^{-1} when the distance between plate and system was increased from 50 mm to 140 mm. This relation can be modelled through the power function, $y = (3.1 \times 10^{8})\, x^{-4.805}$. The model had correlation coefficient of 0.9996, which implies a very

[7] Saslow, W. M. "Maxwell's Theory of Eddy Currents in Thin Conducting Sheets." *American Journal of Physics*, vol. 60, no. 8, 1992, pp. 693–711.

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strong relation between distance and decay constant. The results deviate from theory, as theory suggests the exponential term to be proximally -6, whereas the experiment produced -4.805. This is due to edge effects, non-dipole fields and geometric differences.

Pili (2020)[8] showcased a similar experiment, which experimented the magnetic damping in pendulums. It proved that eddy current damping follows underdamped oscillations. Gonzalez and Alfredo Bol (2006)[9] experimented differing magnetic field strengths at discrete distance intervals. It showed a positive correlation between damping and field strength; however, the experimental method focussed on breaking of an aluminium wheel.

11. Evaluation

11.1. Strengths of experiment

- The controlled trial quantified the decay constant of air resistance. If measuring directly, the graph would have been offset by a certain value. This would make the interpretation of the decay constant ambiguous, as the direct connection between magnetic damping and distance couldn't be analysed. Thus, this led to a very high correlation.
- The use of "Tracker" was important, as it allowed precise markings for $A_{1/2}$. Using a stopwatch would be highly inaccurate due to human reaction. Therefore, "Tracker" significantly reduced uncertainty in time measurements.

11.2. Weaknesses and limitations

Type of error	Error	Impact of error	Improvement
Systematic errors	Edge effect of aluminium plate	The theoretical model functions under the assumption that the aluminium sheet is infinitely large. This would allow eddy current generation throughout the sheet and would have much stronger damping.	Utilise a larger aluminium plate, which approximately covers 3x the entire amplitude of the pendulum. This minimises edge effects, increasing damping.

[8] Pili, Unofre. (2020). Modeling damped oscillations of a simple pendulum due to magnetic braking. Physics Education. 55. 035025. 10.1088/1361-6552/ab7c81.
[9] González, Manuel & Bol-Arreba, A.. (2006). Controlled damping of a physical pendulum: Experiments near critical conditions. European Journal of Physics. 27. 257. 10.1088/0143-0807/27/2/008.

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		However, since the sheet is finite in surface area, the magnetic field lines extend beyond the plate, reducing damping.	
	Non-ideal magnet	The experiment functions under the assumption that the magnetic field provided comes from a dipole source, which implies no physical size. However, at smaller distances, this assumption falters and invalidates the $B \propto \frac{1}{d^3}$ assumption.	A much smaller and spherical magnet could be used to better emulate a dipole source. The magnet must however be dense enough to perform simple harmonic motion.
	Misaligned path	Since the oscillations never truly remain parallel to the plate, the system experiences increased damping at maximas that lie closer to the plate. And since field strength doesn't linearly scale with distance, the experiment's reliability reduces.	Utilise a levelling device to ensure the platform remains level. Ensure the plate is perpendicular to the surface but using a protractor. To ensure parallel swing, use a string to track any deviations in path.
Random Error	Inconsistent release	The pendulum may have controlled initial angle. But while manually releasing, often more energy is added to the system, which alters initial angle.	Use a mechanised remote gate to have consistent releases. This can be done by creating a gateway and opening using a string.
	Large distance inaccuracy	At larger distances, the decay constant is highly difficult to quantify. "Tracker" is not sufficient to gauge this, as demonstrated by the negative uncertainty for d = 140 mm.	Using sensors such as hall effect sensors will allows measurement of B precisely. This will make suer that the field strength does not fall below a threshold value.

12. Further extension

Magnetic braking is a phenomenon commonly used in recent times. Its most versatile use is in regenerative braking in electric vehicles, where kinetic energy of wheels is converted to an electric current which allows energy to be conserved and for the vehicles efficiency to drastically improve. Additionally, such braking is also applicable in industrial braking such as in roller coasters where eddy current generation can be used as a contactless braking where the wear on the braking components over time can be reduced when compared with friction brakes.

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